

Hardware and Materials Guide

Use Batch 1 only. The public Batch 1 pair is Core rev4.1 plus HAT rev4.1 / R2024.12.06. If the board or packaging says Satellite1.1, rev5.1 Core, rev6.1 HAT, or requires an external Wi-Fi antenna, stop: that hardware is unsupported.

ID	Category	Item	Exact specification	Qty	Required
A01	active driver	Dayton Audio ND91-4	4 ohm full-range driver; manufacturer model ND91-4	1	yes
A02	passive radiator	SB Acoustics SB12PACR-00	4 inch aluminum passive radiator with M6 mass post	2	yes
E01	official electronics	FutureProofHomes Satellite1 Batch 1 development kit	Core rev4.1 + HAT rev4.1 / R2024.12.06; not Satellite1.1	1	yes
H01	insert	CNC Kitchen M3 x 5.7 heat-set insert	M3x0.5 internal thread, 5.7 mm length, 4.6 mm maximum OD	48	yes; includes four spares
H02	speaker cable	2-pin JST-XH 2.54 mm speaker lead	22 AWG stranded red/black, each insulated conductor OD <=1.8 mm, 350 mm minimum	1	yes
H03	speaker terminals	2.8 mm fully insulated female quick-disconnects	for 22-18 AWG wire; verify fit on the purchased ND91-4 before crimping	2	recommended; direct solder is acceptable
B01	ballast	mild-steel plate	100 x 112 x 5 mm, edges deburred, dry, light oil removed	2	yes
B02	radiator tuning	self-adhesive tuning mass, lead-free strip	trimmed to 0.78 g per radiator, the two sets matched within 0.02 g; applied centred on the radiator mass post. A single M6 washer is about 1.0 g, which is coarser than the whole target, so threaded mass cannot hit this figure	2 matched sets	yes; final mass requires physical tuning
G00	gasket stock	closed-cell EPDM foam sheet	2.0 mm nominal, soft, smooth skin, ASTM D1056 2A1 or equivalent	one 300 x 300 mm sheet	yes
D01	optional acoustic material	polyester acoustic batting	not installed in RC1; reserve for measurement-led development only	0	no
O01	official printed part	official_mid_plate	ASA (PETG alternative); exact file OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_mid_plate.stl; preserved official STL	1	yes

ID	Category	Item	Exact specification	Qty	Required
O02	official printed part	official_mid_plate_th reads	ASA (PETG alternative); exact file OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_mid_plate_threads.stl; preserved official STL	1	yes
O03	official printed part	official_pcb_spacer	ASA (PETG alternative); exact file OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_pcb_spacer.stl; preserved official STL	1	yes
O04	official printed part	official_lock_ring	ASA (PETG alternative); exact file OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_lock_ring.stl; preserved official STL	1	yes
O05	official printed part	official_top_plate	ASA (PETG alternative); exact file OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_top_plate.stl; preserved official STL	1	yes
O06	official printed part	official_top_plate_snap_in_diffuser_ring	ASA (PETG alternative); exact file OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_top_plate_snap_in_diffuser_ring.stl; preserved official STL	1	yes
O07	official printed part	official_top_plate_mm_buttons	ASA (PETG alternative); exact file OFFICIAL_PARTS/OPTIONAL_MULTIMATERIAL/official_top_plate_mm_buttons.stl; preserved official STL	1	optional alternative
O08	official printed part	official_top_plate_mm_diffuser_ring	ASA (PETG alternative); exact file OFFICIAL_PARTS/OPTIONAL_MULTIMATERIAL/official_top_plate_mm_diffuser_ring.stl; preserved official STL	1	optional alternative
F01	fastener	M3 x 6 socket cap screw	ISO 4762, A2-70 stainless, M3x0.5	4	yes
F02	fastener	M3 x 8 button head screw	ISO 7380-1, A2-70 stainless, M3x0.5	4	yes
F03	fastener	M3 x 8 socket cap screw	ISO 4762, A2-70 stainless, M3x0.5	8	yes
F04	fastener	M3 x 10 socket cap screw	ISO 4762, A2-70 stainless, M3x0.5	4	yes

ID	Category	Item	Exact specification	Qty	Required
F05	fastener	M3 x 10 socket cap screw	ISO 4762, A2-70 stainless, M3x0.5	8	yes
F06	fastener	M3 x 6 button head screw	ISO 7380-1, A2-70 stainless, M3x0.5	4	yes
F07	fastener	M3 x 10 socket cap screw	ISO 4762, A2-70 stainless, M3x0.5	4	yes
F08	fastener	M3 x 8 button head screw	ISO 7380-1, A2-70 stainless, M3x0.5	4	yes
F09	fastener	M3 x 8 button head screw	ISO 7380-1, A2-70 stainless, M3x0.5	4	yes
F10	fastener	M3 x 8 socket cap screw	ISO 4762, A2-70 stainless, M3x0.5	4	yes
F11	fastener	M3 x 8 socket cap screw	ISO 4762, A2-70 stainless, M3x0.5	4	yes

All purchasing availability and prices must be checked by the builder. Manufacturer geometry and electrical parameters are DERIVED_FROM_MANUFACTURER_DRAWING; supplier availability is an ENGINEERING_ESTIMATE.

No structural glue is used. No damping material is installed in RC1. Gaskets are replaceable mechanically compressed EPDM, and the cable seal is TPU 95A.

Source commit 00b8318db6d6. No physical validation has been performed.